Project Background

Tiny code (SQL) - Fine-Tuning Lightweight Language Models for BigQuery SQL Generation

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Project Background (AI590)

Background on Text-to-SQL Generation and Parameter-Efficient Fine-Tuning of Large Language Models

Translating natural language queries into executable SQL statements (Text-to-SQL or NL2SQL) has long been pursued to lower barriers to relational database access for non-expert users. Early approaches relied on rule-based parsers or traditional machine learning techniques. For example, Zhong et al. (2017) introduced Seq2SQL, a reinforcement-learning-based seq2seq model trained to generate structured queries directly from natural language on benchmarks such as WikiSQL. Subsequent pre-large-language-model (PLM) methods, such as RESDSQL, employed encoder-decoder architectures with schema-linking techniques to handle complex multi-table joins and achieve strong results on the Spider benchmark with lower inference costs than pure LLM prompting.

With the rise of large language models (LLMs), two dominant paradigms emerged: in-context prompting and supervised fine-tuning. Prompting methods (e.g., DIN-SQL) decompose the task into sub-problems and provide few-shot examples or schema context within the input prompt, but they suffer from high token consumption and latency on complex queries. Surveys such as Liu et al. (2025) and Hong et al. (2025) document that, when labeled training data are available,

supervised fine-tuning consistently outperforms prompting by adapting model parameters to better capture schema grounding, dialect-specific syntax, and reasoning patterns required for accurate SQL generation.

To make fine-tuning computationally tractable for billion-parameter models, parameter-efficient fine-tuning (PEFT) techniques were developed. Hu et al. (2021) introduced **Low-Rank Adaptation (LoRA)**, which freezes the pre-trained weights of a Transformer and injects trainable low-rank decomposition matrices (rank $r \ll d$) into each layer. Compared with full fine-tuning of GPT-3 175B using Adam, LoRA reduces the number of trainable parameters by up to 10,000× and GPU memory requirements by 3× while matching or exceeding full fine-tuning performance on tasks including GPT-2 and GPT-3 code generation. Dettmers et al. (2023) extended this to **QLoRA** by combining 4-bit quantization (NF4) with LoRA, enabling fine-tuning of 65B-parameter models on a single 48 GB GPU without sacrificing downstream accuracy.

These PEFT methods have proven particularly well-suited to Text-to-SQL because the task benefits from domain-specific adaptation (schema understanding, dialect rules, and value grounding) while requiring far less data and compute than full retraining. Google's official Gemma fine-tuning documentation (2026) explicitly demonstrates this workflow: the philschmid/gretel-synthetic-text-to-sql dataset (a high-quality synthetic corpus of >100k natural-language-to-SQL pairs with schema, reasoning traces, and SQL outputs) is formatted into conversational templates and used to train Gemma models via QLoRA + Hugging Face TRL's SFTTrainer. The guide highlights that this approach enables production-ready SQL generation on modest hardware while preserving the base model's broad knowledge.

Multiple independent projects have applied the same LoRA/QLoRA + Gemma + Gretel combination. Community models on Hugging Face include:

- parole-study-viper/gemma3-text-to-sql — a LoRA adapter (rank 16, target modules covering all attention and MLP layers) on Gemma-3-27B-it, trained on synthetic and curated text-to-SQL data, reported to generate clean, schema-aware queries with joins, aggregations, and filters.

- litert-community/gemma3-1b-ft-text-to-sql and Yuk050/gemma-3-1b-text-to-sql-model — both QLoRA-fine-tuned on the Gretel synthetic dataset, explicitly created for efficient text-to-SQL inference on resource-constrained systems.

Commercial efforts follow a parallel trajectory. Defog's SQLCoder family (7B–70B variants) was built by fine-tuning CodeLlama on large text-to-SQL corpora; subsequent work has further improved these models with LoRA adapters and reinforcement learning from execution feedback, raising exact-match scores from 31.49 % to 49.83 % on benchmarks while reducing syntactic errors. These models have been shown to outperform GPT-3.5 and approach GPT-4 on domain-specific SQL tasks.

Collectively, these surveys, foundational PEFT papers, official framework guides, open-source adapters, and commercial deployments demonstrate that LoRA/QLoRA-based fine-tuning of instruction-tuned LLMs (particularly Gemma-series models) on synthetic text-to-SQL datasets has become a mature, widely validated method for building high-accuracy, dialect-aware SQL generators. The approach addresses the core challenges of schema linking and query complexity while remaining feasible on single-GPU infrastructure — precisely the characteristics that make it appropriate for applied Text-to-SQL projects in 2025–2026.